Factory Method

Create Email/SMS/Push notifiers without client coupling

Season 1 · GoF · Beginner

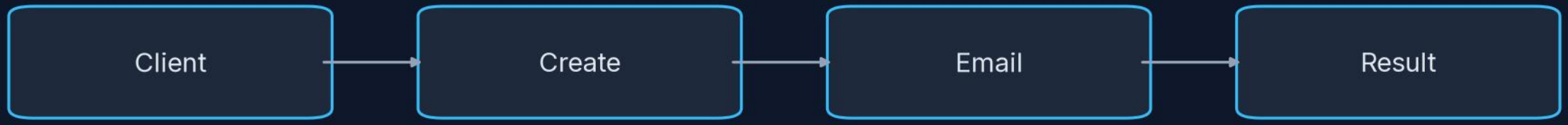
Today's Hook

- Problem: Create Email/SMS/Push notifiers without client coupling
- Naive pain cue: caught=unknown channel fax
- We will refactor to Factory Method with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Create Email/SMS/Push notifiers without client coupling
- Constraints: solve «Create Email/SMS/Push notifiers without client coupling» with Factory Method — keep the demo runnable and the failure path observable.
- Naive pain: Naive: client switches on channel strings and news concrete types.
- Success looks like: email:a@b.c:hi

Structure



Factory Method: Create Email/SMS/Push notifiers without client coupling

Real-World Scenario

- Running example: Create Email/SMS/Push notifiers without client coupling
- Key types: EmailNotifier, PushNotifier
- Happy-path stdout cue: email:a@b.c:hi
- Failure-path stdout cue: caught=unknown channel fax

Naive Code

```java
package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;

/** Naive: client switches on channel strings and news concrete types. */
public final class NaiveNotify {
  public static String send(String channel, String to, String body) {
    if ("email".equals(channel)) return "email:" + to + ":" + body;
    if ("sms".equals(channel)) return "sms:" + to + ":" + body;
    if ("push".equals(channel)) return "push:" + to + ":" + body;
    throw new IllegalArgumentException("unknown channel " + channel);
  }
}
```

Why Naive Hurts

- Naive: client switches on channel strings and news concrete types.
- Without Factory Method, call sites absorb the wrong details.
- Failure mode to show on camera: caught=unknown channel fax
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;
public record EmailNotifier() implements Notifier {
  public String send(String to, String body) { return "email:" + to + ":" + body; }
}


package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;
public record PushNotifier() implements Notifier {
  public String send(String to, String body) { return "push:" + to + ":" + body; }
}
```

Runnable main

```java
package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;

public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== Factory Method: Email/SMS/Push notifiers ===");
    System.out.println("--- naive ---");
    System.out.println(NaiveNotify.send("email", "a@b.c", "hi"));
    try { NaiveNotify.send("fax", "a", "x"); } catch (IllegalArgumentException e) { System.out.println("caught=" + e.getMessage()); }
    System.out.println("--- factory method ---");
    NotifierFactory email = new EmailFactory();
    NotifierFactory sms = new SmsFactory();
    System.out.println(email.notifyUser("a@b.c", "hi"));
    System.out.println(sms.notifyUser("+1", "hi"));
    System.out.println(new PushFactory().notifyUser("device", "hi"));
  }
}
```

Walkthrough Output

```
=== Factory Method: Email/SMS/Push notifiers ===
--- naive ---
email:a@b.c:hi
caught=unknown channel fax
--- factory method ---
email:a@b.c:hi
sms:+1:hi
push:device:hi
```

Step-by-Step Trace

- Match stdout to stages — look for: email:a@b.c:hi
- Then the injected failure: caught=unknown channel fax
- Re-run: java -cp ... 04-factory-method ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;
public record EmailNotifier() implements Notifier {
  public String send(String to, String body) { return "email:" + to + ":" + body; }
}


package com.buildwithwaqar.patterns.gof.creational.factorymethod.problem;
public record PushNotifier() implements Notifier {
  public String send(String to, String body) { return "push:" + to + ":" + body; }
}
```

Trade-offs

- Use Factory Method when «Create Email/SMS/Push notifiers without client coupling» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Factory Method in product language.

Related Patterns

- Name the pattern usually confused with Factory Method and separate the forces.
- Sketch the roles for «Create Email/SMS/Push notifiers without client coupling» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Create Email/SMS/Push notifiers without client coupling
- Naive pain → Factory Method boundary → Demo stdout.
- Remember the failure cue: caught=unknown channel fax
- Break a ProblemTest assertion, watch it fail, restore it.